

LAB EXPERIMENT RECORD

Set 1: Monthly Sales Data

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	DIVYASREE.M
Register Number	192524098
Date	29 September 2026
Day	Tuesday

Aim

To visualize monthly sales data using a line chart, bar chart and scatter plot, and to build an interactive dashboard combining the line and bar charts using R and Shiny.

Dataset: Monthly Sales

Month	Sales (\$)
January	15000
February	18000
March	22000
April	20000
May	23000

Procedure

1. Create the month and sales vectors in R.
2. Use `plot()` with `type = "o"`, `axis()` and `grid()` to draw the labelled line chart.
3. Use `barplot()` with `names.arg` to compare sales across months.
4. Create an advertising budget vector of the same length, draw the scatter plot with `plot()`, and add a regression line using `lm()` and `abline()`.
5. Install and load the shiny package (`install.packages("shiny")`), then create the UI with `fluidPage()` and the server function.
6. Run the app with `shinyApp()` and use the month and colour selectors to explore both charts interactively.

Key R Functions Used

Function	Purpose
<code>plot()</code> / <code>axis()</code> / <code>grid()</code>	Draw line and scatter plots, add month labels and grid lines.
<code>barplot()</code>	Creates bar charts.
<code>lm()</code> / <code>abline()</code> / <code>cor()</code>	Fit and draw the regression line; measure correlation.
<code>fluidPage()</code> / <code>sidebarLayout()</code> / <code>selectInput()</code>	Build the Shiny layout and month/colour controls.
<code>reactive()</code> / <code>renderPlot()</code> / <code>renderText()</code>	Filter data and redraw plots and summary dynamically.
<code>shinyApp()</code>	Runs the interactive Shiny application.

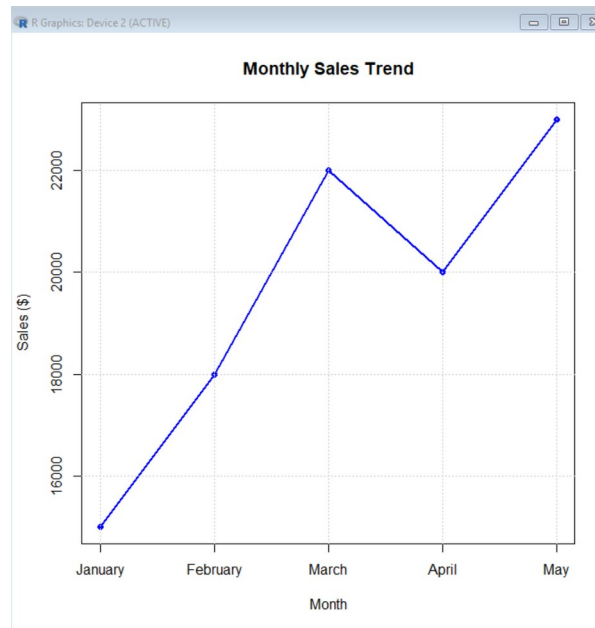
PROGRAM 1: Line Chart of Monthly Sales

Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

```
# Dataset
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

# Line Chart
plot(1:5, sales,
     type = "o",
     col = "blue",
     lwd = 2,
     pch = 16,
     xaxt = "n",
     xlab = "Month",
     ylab = "Sales ($)",
     main = "Monthly Sales Trend")

axis(1, at = 1:5, labels = month)
grid()
```



Output 1: Monthly Sales Trend (Line Chart)

Observation: Sales rise steadily from \$15,000 in January to \$22,000 in March, dip to \$20,000 in April, and reach the highest value of \$23,000 in May. Overall the trend is upward.

PROGRAM 2: Bar Chart of Monthly Sales

Generate a bar chart to display the top-selling months for the year. Label the chart elements.

```
# Dataset
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

# Bar Chart
barplot(sales,
        names.arg = month,
        col = "skyblue",
        xlab = "Month",
        ylab = "Sales ($)",
        main = "Monthly Sales")
```



Output 2: Monthly Sales (Bar Chart)

Observation: May is the top-selling month (\$23,000), followed by March (\$22,000). January has the lowest sales (\$15,000). Total sales for the period are \$98,000 with an average of \$19,600 per month.

PROGRAM 3: Scatter Plot of Advertising Budget vs Sales

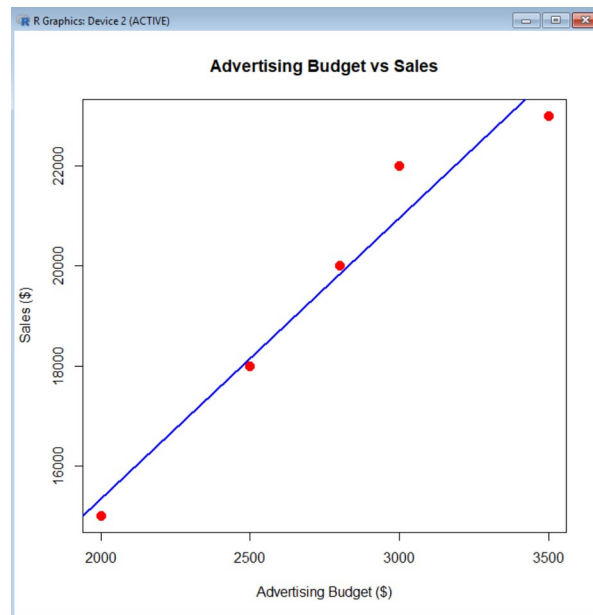
Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.

```
# Dataset
sales      <- c(15000, 18000, 22000, 20000, 23000)
advertising <- c(2000, 2500, 3000, 2800, 3500)

# Scatter Plot
plot(advertising, sales,
     pch = 19,
     col = "red",
     cex = 1.5,
     xlab = "Advertising Budget ($)",
     ylab = "Sales ($)",
     main = "Advertising Budget vs Sales")

# Regression (trend) line
model <- lm(sales ~ advertising)
abline(model, col = "blue", lwd = 2)

# Strength of relationship
cor(advertising, sales)      # 0.977
summary(model)$r.squared    # 0.955
```



Output 3: Advertising Budget vs Sales (Scatter Plot)

Insight: There is a strong positive relationship between advertising budget and sales (correlation $r = 0.98$). The fitted line ($\text{Sales} \approx 4125 + 5.61 \times \text{Budget}$, $R^2 = 0.96$) shows that every additional \$1 spent on advertising is associated with about \$5.6 more in sales. March lies slightly above the line and April slightly below it, but all points follow the upward trend.

Note: The given dataset contains only Month and Sales, so the advertising budget values (2000, 2500, 3000, 2800, 3500) were assumed for analysis. With only five points, the result shows association, not proof of cause and effect.

PROGRAM 4: Interactive Sales Dashboard

Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

```
library(shiny)

# Dataset
sales_data <- data.frame(
  Month = factor(c("January", "February", "March", "April", "May"),
    levels = c("January", "February", "March", "April", "May")),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)

ui <- fluidPage(
  titlePanel("Monthly Sales Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      selectInput("months", "Select Months:",
        choices = levels(sales_data$Month),
        selected = levels(sales_data$Month),
        multiple = TRUE),
      selectInput("color", "Chart Colour:",
        choices = c("blue", "darkgreen", "purple", "red")),
      hr(),
      strong("Summary"),
      verbatimTextOutput("summary")
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("linePlot", height = "350px")),
        column(6, plotOutput("barPlot", height = "350px"))
      )
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({
    sales_data[sales_data$Month %in% input$months, ]
  })

  output$linePlot <- renderPlot({
    d <- filtered()
    req(nrow(d) > 0)
    plot(seq_len(nrow(d)), d$Sales, type = "o", pch = 16, lwd = 2,
      col = input$color, xaxt = "n",
      xlab = "Month", ylab = "Sales ($)", main = "Monthly Sales Trend")
    axis(1, at = seq_len(nrow(d)), labels = as.character(d$Month))
    grid()
  }, width = 420, height = 340)

  output$barPlot <- renderPlot({
    d <- filtered()
    req(nrow(d) > 0)
    barplot(d$Sales, names.arg = as.character(d$Month), col = input$color,
      xlab = "Month", ylab = "Sales ($)", main = "Monthly Sales")
  }, width = 420, height = 340)

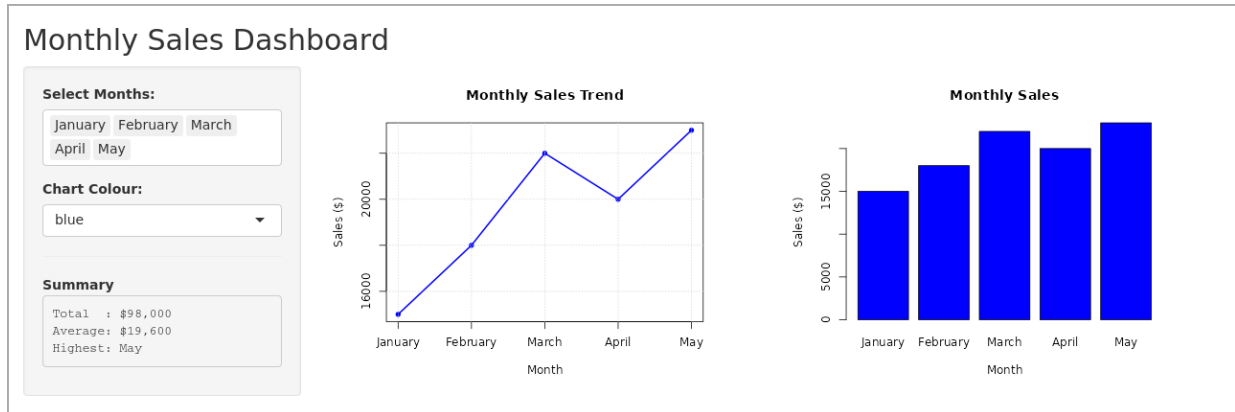
  output$summary <- renderText({
    d <- filtered()
    req(nrow(d) > 0)
    paste0("Total : $", format(sum(d$Sales), big.mark = ","), "\n",
```

```

    "Average: $", format(round(mean(d$Sales)), big.mark = ","), "\n",
    "Highest: ", d$Month[which.max(d$Sales)]
  })
}

shinyApp(ui = ui, server = server)

```



Output 4: Interactive Monthly Sales Dashboard (R Shiny)

Observation: The dashboard shows the line chart and bar chart side by side. Selecting or removing months updates both charts and the summary panel (total, average and highest month) instantly, and the colour selector restyles both charts. With all months selected, total sales are \$98,000, the average is \$19,600 and May is the highest month.

Result

The monthly sales data was successfully visualized using line, bar and scatter charts in R, and an interactive Shiny dashboard combining the line and bar charts was created and executed.